

## PTEN loss in glioma cell lines leads to increased extracellular vesicles biogenesis and PD-L1 cargo in a PI3K-dependent manner.

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Phosphatase and Tensin Homologue (PTEN) is one of the most frequently lost tumor suppressors in cancer and the predominant negative regulator of the PI3K/AKT signaling axis. A growing body of evidence has highlighted the loss of PTEN with immuno-modulatory functions including the upregulation of the programmed death ligand-1 (PD-L1), an altered tumor derived secretome that drives an immunosuppressive tumor immune microenvironment (TIME), and resistance to certain immunotherapies. Given their roles in immunosuppression and tumor growth, we examined whether the loss of PTEN would impact the biogenesis, cargo, and function of extracellular vesicles (EVs) in the context of the anti-tumor associated cytokine interferon- $\gamma$  (IFN- $\gamma$ ). Through genetic and pharmacological approaches, we show that PD-L1 expression is regulated by JAK/STAT signaling, not PI3K signaling. Instead, we observe that PTEN loss positively upregulates cell surface levels of PD-L1 and enhances the biogenesis of EVs enriched with PD-L1 in a PI3K-dependent manner. We demonstrate that because of these changes, EVs derived from glioma cells lacking PTEN have a greater ability to suppress T cell receptor (TCR) signaling. Taken together, these findings provide important new insights into how the loss of PTEN can contribute to an immunosuppressive TIME, facilitate immune evasion, and highlight a novel role for PI3K signaling in the regulation of EV biogenesis and the cargo they contain.

## PTEN loss in glioma cell lines leads to increased extracellular vesicle biogenesis and PD-L1 cargo in a PI3K-dependent manner.

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## The PTEN-associated immune prognostic signature reveals the landscape of the tumor microenvironment in glioblastoma.

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Glioblastoma (GBM) is a common brain tumor with a complex and diverse tumor microenvironment (TME). As PTEN mutation is the most common mutation in GBM, we aimed to investigate how PTEN mutation regulates the immune response in GBM TME and thus affects the prognosis of GBM patients. In this study, we conducted a comprehensive analysis of multiple levels of data, including whole-exome sequencing (WES), transcriptome RNA sequencing, patient survival and immune signatures, to study the relationship between PTEN mutation and TME in GBM. We developed an immune-related prognostic signature (IPS) based on the PTEN-associated immune-related genes (IRGs), and the IPS exhibited a powerful prognosis prediction capacity in different GBM cohorts. A scoring nomogram based on the IPS was also established for clinical application. In addition, the correlations of the IPS with tumor immune cell infiltration and immune checkpoints were systematically analyzed. This study illustrates the influence of PTEN mutation on the immune microenvironment of GBM. Our IPS, which is sensitive to PTEN mutation status, can enhance the prognosis prediction ability for GBM patients and provides potential targets for immunotherapy.

## Resistance to EGF receptor inhibitors in glioblastoma mediated by phosphorylation of the PTEN tumor suppressor at tyrosine 240.

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Glioblastoma multiforme (GBM) is the most aggressive of the astrocytic malignancies and the most common intracranial tumor in adults. Although the epidermal growth factor receptor (EGFR) is overexpressed and/or mutated in at least 50% of GBM cases and is required for tumor maintenance in animal models, EGFR inhibitors have thus far failed to deliver significant responses in GBM patients. One inherent resistance mechanism in GBM is the coactivation of multiple receptor tyrosine kinases, which generates redundancy in activation of phosphoinositide-3'-kinase (PI3K) signaling. Here we demonstrate that the phosphatase and tensin homolog deleted on chromosome 10 (PTEN) tumor suppressor is frequently phosphorylated at a conserved tyrosine residue, Y240, in GBM clinical samples. Phosphorylation of Y240 is associated with shortened overall survival and resistance to EGFR inhibitor therapy in GBM patients and plays an active role in mediating resistance to EGFR inhibition in vitro. Y240 phosphorylation can be mediated by both fibroblast growth factor receptors and SRC family kinases (SFKs) but does not affect the ability of PTEN to antagonize PI3K signaling. These findings show that, in addition to genetic loss and mutation of PTEN, its modulation by tyrosine phosphorylation has important implications for the development and treatment of GBM.

## Progress in targeting PTEN/PI3K/Akt axis in glioblastoma therapy: Revisiting molecular interactions.

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Glioblastoma (GBM) is one of the most malignant cancers of central nervous system and due to its sensitive location, surgical resection has high risk and therefore, chemotherapy and radiotherapy are utilized for its treatment. However, chemoresistance and radio-resistance are other problems in GBM treatment. Hence, new therapies based on genes are recommended for treatment of GBM. PTEN is a tumor-suppressor operator in cancer that inhibits PI3K/Akt/mTOR axis in diminishing growth, metastasis

and drug resistance. In the current review, the function of PTEN/PI3K/Akt axis in GBM progression is evaluated. Mutation or depletion of PTEN leads to increase in GBM progression. Low expression level of PTEN mediates poor prognosis in GBM and by increasing proliferation and invasion, promotes malignancy of tumor cells. Moreover, loss of PTEN signaling can result in therapy resistance in GBM. Activation of PTEN signaling impairs GBM metabolism via glycolysis inhibition. In contrast to PTEN, PI3K/Akt signaling has oncogenic function and during tumor progression, expression level of PI3K/Akt enhances. PI3K/Akt signaling shows positive association with oncogenic pathways and its expression similar to PTEN signaling, is regulated by non-coding RNAs. PTEN upregulation and PI3K/Akt signaling inhibition by anti-cancer agents can be beneficial in interfering GBM progression. This review emphasizes on the signaling networks related to PTEN/PI3K/Akt and provides new insights for targeting this axis in effective GBM treatment.

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## Implications for immunotherapy of tumor-mediated T-cell apoptosis associated with loss of the tumor suppressor PTEN in glioblastoma.

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The ability of glioma cells to escape the immune system remains a significant barrier to successful immunotherapy. Here we demonstrate that loss of the PTEN tumor suppressor gene, with associated activation of the PI3K/Akt/mTOR pathway, leads to a human glioma phenotype that induces autologous T-cell apoptosis upon contact. The PTEN status of pathologically confirmed glioblastoma specimens was defined, and primary cultures established after surgical resection of tumor from 26 patients. Autologous T-cells were isolated from these patients, and after T-cell activation was induced, these cells were co-cultured with matched autologous glioma cells, either alone, or after treatment with one of three inhibitors of the PI3K/Akt/mTOR pathway. When co-cultured with autologous T-cells, PTEN wild-type tumor cells induced apoptosis in a minimal number of activated T-cells (6-12% of T-cells), whereas tumors with PTEN loss induced much more profound levels of T-cell apoptosis (42-56% of T-cells). Prior treatment of PTEN-deficient tumor cells with specific inhibitors of the PI3K/Akt/mTOR pathway diminished T-cell apoptosis to levels seen after co-culture with wild-type PTEN tumor cells, suggesting that PTEN loss confers this immunoresistant phenotype through the PI3K/Akt/mTOR pathway. These results suggest that PTEN-deficient glioblastoma patients are suboptimal candidates for immunotherapy. In addition, our results raise the possibility of combining T-cell based immunotherapy protocols with clinical inhibitors of the PI3K/Akt/mTOR pathway.

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## PTEN Mutations Trigger Resistance to Immunotherapy.

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Immune checkpoint-blocking antibodies are actively used to treat multiple cancer types; however, the underlying resistance mechanism remains unclear. In a recent study, Zhao et al. (Nat. Med. 2019;25:462-469) found that somatic PTEN mutations were associated with resistance to immune checkpoint inhibitors by altering immunosuppressive environments in patients with glioblastomas.

## Genomic Approaches to Understanding Response and Resistance to Immunotherapy.

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Immunotherapy has led to a paradigm shift in the treatment of some malignancies, providing long-term, durable responses for patients with advanced cancers. However, such therapy has benefited only a subset of patients, with some patients failing to respond to treatment at all and others achieving a limited response followed by tumor progression. Understanding factors contributing to an effective response and further elucidating mechanisms of resistance will be crucial as these therapies are applied more broadly. Genomics-based approaches have significantly advanced the study of response and resistance to immunotherapy in general, and to immune checkpoint blockade more specifically. Here, we review how genomic and transcriptomic approaches have identified both somatic and germline positive correlates of response, including high mutational/neoantigen load and low intratumoral heterogeneity, among others. The genomic analysis of resistant tumors has additionally identified crucial factors involved in resistance to immune checkpoint blockade, including loss of PTEN and upregulation of other immune checkpoints. Overall, the continued use of genomic techniques at the point of care, combined with appropriate functional studies, would ideally lead to a better understanding of why certain patients respond to immune-based therapies, allowing clinicians to identify the subset of patients likely to benefit from such therapy, and potentially providing insight into how other therapies may be added in combination to increase the number of patients who may benefit from immunotherapy. Clin Cancer Res; 22(23); 5642-50. ©2016 AACR.

## The novel Hsp90 inhibitor NXD30001 induces tumor regression in a genetically engineered mouse model of glioblastoma multiforme.

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Glioblastoma multiforme (GBM) has an abysmal prognosis. We now know that the epidermal growth factor receptor (EGFR) signaling pathway and the loss of function of the tumor suppressor genes p16Ink4a/p19ARF and PTEN play a crucial role in GBM pathogenesis: initiating the early stages of tumor development, sustaining tumor growth, promoting infiltration, and mediating resistance to therapy. We have recently shown that this genetic combination is sufficient to promote the development of GBM in adult mice. Therapeutic agents raised against single targets of the EGFR signaling pathway have proven rather inefficient in GBM therapy, showing the need for combinatorial therapeutic approaches. An effective strategy for concurrent disruption of multiple signaling pathways is via the inhibition of the molecular chaperone heat shock protein 90 (Hsp90). Hsp90 inhibition leads to the degradation of so-called client proteins, many of which are key effectors of GBM pathogenesis. NXD30001 is a novel second generation Hsp90 inhibitor that shows improved pharmacokinetic parameters. Here we show that NXD30001 is a potent inhibitor of GBM cell growth in vitro consistent with its capacity to inhibit several key targets and regulators of GBM biology. We also show the efficacy of NXD30001 in vivo in an EGFR-driven genetically engineered mouse model of GBM. Our findings establish that the Hsp90 inhibitor NXD30001 is a therapeutically multivalent molecule, whose actions strike GBM at the core of its drivers of tumorigenesis and represent a compelling rationale for its use in GBM treatment.

## Gamma interferon-mediated superinduction of B7-H1 in PTEN-deficient glioblastoma: a

## paradoxical mechanism of immune evasion.

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B7 homolog 1 (B7-H1) is a recently discovered immunoresistance protein that is regulated posttranscriptionally after PTEN loss in malignant glioma, a deadly form of brain tumor. Here, the impact of gamma-interferon-mediated activation of B7-H1 was investigated in glioblastoma patients with PTEN loss. Lymphocytes and T cells were selected for apoptosis assays after 1 : 1 coculture with autologous glioma cells. Gamma interferon treatment of PTEN-deficient tumors resulted in superinduction of B7-H1 protein that correlated with increased T-cell apoptosis, an effect dependent upon activation of the PI3-kinase pathway. The combination of PTEN loss and gamma-interferon exposure in glioblastoma patients results in an exceptionally immunoresistant phenotype that may negate adaptive immunity through induction of T-cell apoptosis.

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